FALL, 2025

INVESTMENT ANALYSIS REPORT



BRAC UNIVERSITY
Inspiring Excellence

FIN433: SECURITY ANALYSIS & PORTFOLIO MANAGEMENT

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Name	ID	Contribution
Sabit Mohammed Walid Adnan		• BRAC Bank Stock Analysis • Efficient Frontier Manual. • Portfolio Construction
Irfanur Rahman		• GP Stock Analysis • Risk-Return & Covariance Matrix • Efficient Frontier Simulation
Elma Taseen		• Square Pharmaceuticals Ltd. Stock Analysis
Sajid Asif		• Apex Tannery Ltd.Analysis

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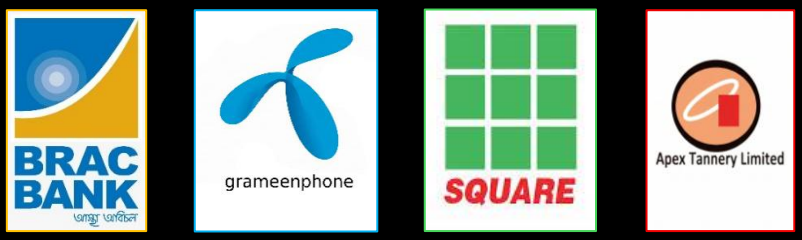
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PART 01: STOCK SELECTION



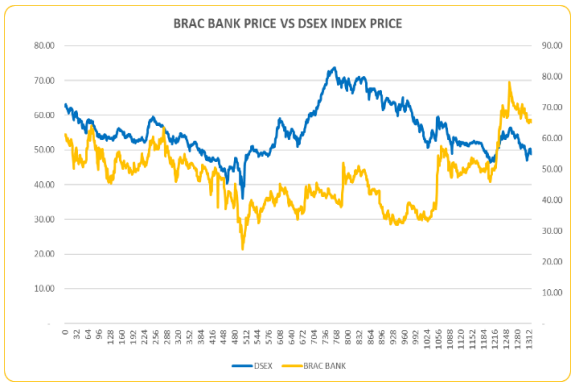


STOCK 01: BRAC BANK PLC.

Dhaka Stock Exchange (DSE: BRACBANK) Sector: Bank As of 31st Dec 2025	Current Price: BDT 63.10 Target Price: BDT 70.74 Upside: 12.1%	Recommendation: Hold

1. EXECUTIVE SUMMARY: BRAC BANK

Recommendation	Hold
Valuation Date	31.13.2025
Implied Price	45.46
Current Price	63.1
Target Price (Forward EPS)	70.74
Upside	12.11%
Shares Outstanding	1,990,926,167
Market Capitalization (mn)	134,586.61



We initiate coverage on BRAC Bank Limited with a HOLD and a 12-month valuation range of BDT 55–71 per share, with a central fair value of BDT 70.7 under our forward-earnings eps. This implies a potential upside of around 12% from the price of BDT 63.10 as of 31st Dec 2025, contingent on the bank delivering its projected earnings growth. Our valuation is primarily anchored on an industry-relative P/E framework and is triangulated using earnings sensitivity scenarios, and peer-based comparative valuation. BRAC Bank is one of Bangladesh's leading private commercial banks, distinguished by its strong image, diversified loan portfolio, & consistent profitability. The bank has demonstrated robust operational momentum, with revenue growth of 39.93% and operating profit growth of 68.18%, both exceeding the respective industry averages of 38.99% and 58.63%. This superior growth profile is further reflected in BRAC Bank's higher ROA and ROE relative to peers, justifying its premium market valuation.

BRAC BANK IS A RETURN LEADER – DRIVEN BY ROA & ROE LEADERSHIP

Description	BRAC Bank	Industry Avg.
ROE	17.43%	15.99%
ROA	1.46%	1.54%
Revenue Growth	39.99%	38.99%
Operating Profit Growth	68.18%	58.63%

BRAC Bank distinguishes itself within Bangladesh's banking sector as a return-focused outperformer, consistently delivering return on assets (ROA) and return on equity (ROE) above the industry average. With a ROA of 1.46% and a ROE of 17.43%, the bank demonstrates strong efficiency in generating earnings from both its assets and shareholder capital, setting it apart from peers. These return metrics are central to the bank's investment appeal, as they directly drive long-term shareholder value and profitability sustainability.

The bank's superior returns are reinforced by robust operational momentum. BRAC Bank achieved revenue growth of 39.99% and operating profit growth of 68.18%, both above the industry averages

of 38.99% and 58.63%, respectively. The fact that operating profit growth outpaced revenue growth reflects improving operating leverage, where earnings expand faster than the bank's asset base. This dynamic contributes directly to higher ROA and explains why BRAC Bank maintains structurally stronger profitability metrics than the broader banking sector. High ROE further underscores BRAC Bank's efficiency in deploying shareholder capital. The bank's ability to convert earnings growth into shareholder returns indicates disciplined capital allocation and provides resilience against potential credit or macroeconomic shocks. Sustained outperformance in ROA and ROE supports the bank's capacity to deliver consistent, high-quality earnings growth.

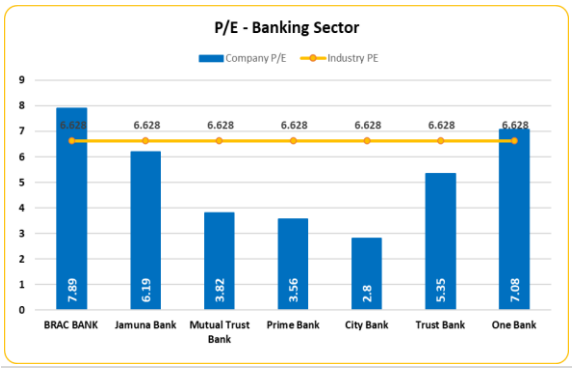
Looking ahead, BRAC Bank appears positioned for continued return-led earnings acceleration. The bank's revenue and operating profit momentum, coupled with above-industry ROA and ROE, suggests that it can maintain profitable growth without relying on balance-sheet expansion alone. These return advantages form the foundation for both earnings resilience and long-term shareholder value creation.

In our view, BRAC Bank is best characterized as a profitability-led compounder rather than a balance-sheet or value play. Investors seeking exposure to a banking franchise with demonstrable ROA and ROE superiority, supported by strong earnings momentum, should view BRAC Bank as a high-quality core holding, while closely monitoring the persistence of its return advantage as the primary determinant of long-term performance.

BRAC BANK: VALUATION INSIGHTS THROUGH P/E AND P/B ANALYSIS

Description	BRAC Bank	Industry Avg.
Revenue Growth	7.89	6.63
Operating Profit Growth	1.22	0.68

BRAC Bank is one of the leading private commercial banks in Bangladesh, widely recognized for its strong retail and SME banking operations, as well as its consistent financial performance. As of December 31, 2025, the bank's shares are trading at BDT 63.10 on the Dhaka Stock Exchange. Evaluating its investment potential requires examining both earnings and book value-based valuation metrics, along with sensitivity analyses to understand how different market assumptions affect intrinsic value.



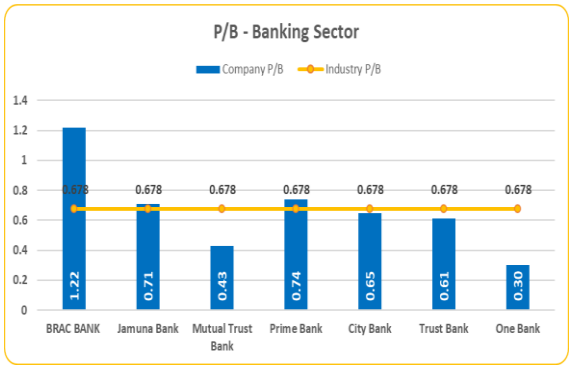
Starting with earnings-based valuation, the Price-to-Earnings (P/E) ratio provides a useful lens to assess whether BRAC Bank's current market price reflects its earnings potential. The industry average P/E ratio stands at 6.63, while BRAC Bank's audited earnings per share (EPS) are BDT 6.86. Using this industry multiple, the intrinsic value of BRAC Bank's stock

calculates to BDT 45.46. Comparing this intrinsic value to the market price of BDT 63.10 reveals that the stock is trading at a 42.83% premium.

BRAC Bank Stock Price Sensitivity Analysis (P/E Based)

		Industry Average P/E Ratio			
		5.00	6.63	7.00	8.00
Audited EPS	6.86	45.46	45.46	45.46	45.46
TTM EPS	7.89	39.45	52.29	55.23	63.12
Forward EPS	10.67	53.35	70.72	74.69	85.36

A deeper look through a P/E-based sensitivity analysis reinforces these findings. If we consider the audited EPS of 6.86 and vary the P/E ratio from 5.0 to 8.0, the corresponding intrinsic value ranges from BDT 34.30 at the low end to BDT 54.88 at the high end. Using trailing twelve-month (TTM) EPS of 7.89, the intrinsic value under the same P/E range moves from BDT 39.45 to BDT 63.12. When factoring in forward EPS of 10.67, which reflects expected earnings growth, the intrinsic value could rise significantly, ranging from BDT 53.35 to BDT 85.36, even under average P/E multiples. This indicates that the market is pricing in BRAC Bank's future growth, particularly in its SME and digital banking segments, which offer scalable revenue streams and long-term profitability. While the stock currently trades above historical earnings-based fair value, anticipated growth could justify the premium over time.



Valuation from a book value perspective, which is particularly relevant for banks, BRAC Bank's audited BVPS is BDT 39.21. The industry average Price-to-Book (P/B) ratio is 0.68, which results in an intrinsic value of BDT 26.56. Comparing this with the current market price shows that BRAC Bank is trading at 137.53% premium to book value.

This significant premium reflects the market's recognition of the bank's competitive advantages that are not fully captured in the book value, such as its strong brand equity, extensive branch network, and high-quality loan portfolio.

BRAC Bank Stock Price Sensitivity Analysis (P/B Based)

		Industry Average P/B Ratio			
		0.68	1.00	1.50	2.00
Audited BVPS	39.21	26.56	39.21	58.82	78.42
TTM BVPS	51.73	35.05	51.73	77.60	103.46

A sensitivity analysis of the P/B ratio further illustrates this point. Considering the audited BVPS of 39.21, the intrinsic value ranges from BDT 26.56 at a P/B of 0.68 to BDT 78.42 at a P/B of 2.0. Using TTM BVPS of 51.73, the intrinsic value under the same P/B assumptions ranges from BDT 35.05 to BDT 103.46. This analysis demonstrates that the market is pricing not just the bank's current assets, but also on its future earnings potential, operational efficiencies, and intangible advantages that drive long-term shareholder value.

BRAC Bank's strong earnings growth trajectory is a key factor supporting its market valuation. The bank's audited EPS of 6.86 has already shown steady performance, while its TTM EPS of 7.89 and forward EPS of 10.67 suggest continued momentum in profitability. This earnings strength underpins higher P/E multiples and indicates that the premium reflected in the current market price may be justified by future earnings expansion.

Beyond financial metrics, BRAC Bank's operational strengths reinforce its competitive position. Its diversified product offerings, focus on SME and retail lending, and strong digital banking infrastructure provide scalable growth opportunities. The bank's brand reputation and widespread branch network create a moat that protects its market share and enhances customer loyalty. These qualitative factors explain why stock trades at a premium relative to book value, as traditional book value measures often fail to capture intangible advantages such as brand equity, operational efficiency, and growth potential.

While BRAC Bank's valuation is currently elevated relative to both P/E and P/B-based intrinsic values, the stock offers different implications for investors depending on their strategy. For long-term investors, the forward earnings growth and strong competitive positioning make BRAC Bank an attractive opportunity despite its premium valuation. The market appears to have already priced in the expected expansion, which could lead to continued upside if earnings meet or exceed projections. On the other hand, value-oriented investors may view the stock's current price as relatively high and may prefer to wait for a more favorable entry point, particularly if short-term market fluctuations present temporary corrections.

Risk considerations also warrant attention. Macroeconomic factors, such as interest rate volatility, inflation, regulatory changes, and credit risk in the banking sector, could impact BRAC Bank's performance. Additionally, if the market's optimistic growth expectations fail to materialize, the stock could experience short-term downside pressure given the current premium valuation.

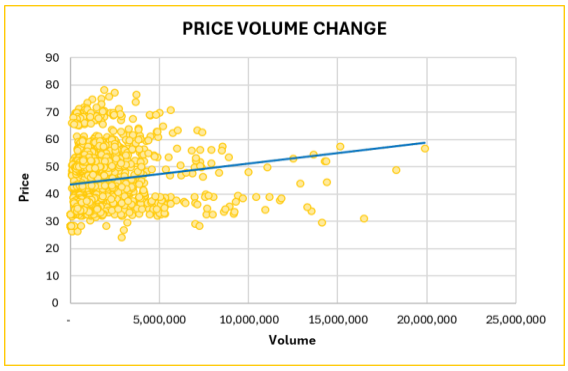
In conclusion, BRAC Bank stands out as a high-quality banking stock in Bangladesh. Its robust earnings, forward growth potential, and operational strengths justify investor interest, though at present, the stock trades at a premium relative to intrinsic valuation metrics. For long-term investors, the bank offers attractive growth prospects and the potential for capital appreciation. For more risk-averse or value-focused investors, careful monitoring and potential timing of entry may be warranted. Overall, BRAC Bank represents a blend of stability, growth, and market recognition, making it a notable player in the Bangladeshi banking sector.

BRAC BANK PRICE VOLUME ANALYSIS



Around 2018, BRAC Bank stocks start near the 60-70 BDT range and enter a significant downtrend over two years, characterized by lower highs and lower lows. Volume during this time is relatively moderate, suggesting a gradual loss of investor confidence or market pressure. Around the beginning of 2020, likely coinciding with the global COVID-19 pandemic shock, the price experiences a severe sharp drop, falling from around 50 to the low 20s. This is a massive loss of value in a very short period. The volume spikes around this time are relatively small compared to later spikes, suggesting the initial selling panic may not have been accompanied by massive institutional unloading, or the scale of the collapse was simply too fast for high volume to register on a cumulative basis.

Following the crash, the stock stages a very strong, high-volume recovery. Volume spikes significantly in 2020 as stocks rebound from the low 20s to the 40-50 range. The high volume suggests strong conviction in the rally. This is the most damning period for sustained performance. After the 2020-2021 stalls, the price enters a prolonged period of stagnation and low volatility in the 30-40 range. Specifically, from mid-2022 to mid-2024, the price is flatlining, predominantly holding steady around 30-35. The volume is low during this period, indicating a lack of investor interest, a 'wait-and-see' attitude, or perhaps a lack of new catalysts for the bank's fundamentals. A stock that trades flat for over two years often suggests a failure to grow or an inability to capture market excitement



The most striking feature of the entire chart is the massive, almost vertical price increase starting in late 2024 and peaking in 2025, taking the stock from the low 30s to almost 80. This is nearly 150% rally. This price surge is accompanied by the highest volume spikes seen in the entire history of the chart (near 20,000,000). High volume on a rally generally confirms the price movement is legitimate and backed by strong demand. The price sharply pulls back from its high near 80 to the mid-60s by the end of the chart's visible data. This creates a potentially concerning "double top" or "blow-off top" pattern. The sudden, extreme volatility and the sharp reversal suggest the rally may have been fueled by speculation or a short-term, unsustainable

catalyst. The current price level (mid-60s) is still high historically, but the sharp sell-off indicates that the momentum has been severely broken, which is often a signal of exhaustion or profit-taking by large investors who initiated the rally.

BRAC BANK TECHNICAL ANALYSIS



BRAC Bank's recent price behavior reflects a phase of consolidation rather than structural deterioration, which is why a hold recommendation is appropriate at this stage. After reaching a peak in early September, the stock entered a controlled correction and has gradually moved lower toward the BDT 63 region. This decline has been orderly, characterized by lower highs and modest price ranges, suggesting profit-taking and risk rebalancing rather than panic-driven selling.

What stands out is how the stock continues to defend the BDT 63 support level. Repeated tests of this zone have resulted in buying interest, as reflected by multiple long lower wicks on the chart. These price reactions indicate that downside pressure is being absorbed by longer-term participants who view current levels as fair value. While the rebounds from this support have not been aggressive, the absence of heavy follow-through selling implies that bears are losing momentum.

Volume dynamics further reinforce the case for holding. Trading activity has tapered off as prices declined, which typically signals seller exhaustion rather than distribution. If institutional investors were exiting positions aggressively, volume would be expanding on down days. Instead, the market appears to be in a wait-and-see mode, consistent with stock transitioning from correction to base formation.

From a broader perspective, BRAC Bank's multi-month and annual performance remains positive, even after the recent pullback. The current weakness therefore appears to be a cyclical correction within a longer-term uptrend, not a trend reversal. Selling at this stage would mean exiting near support, where risk–reward is no longer favorable for liquidation but also not yet compelling enough for fresh accumulation.

Given these factors, holding the stock allows investors to preserve positioning while awaiting confirmation of the next directional move. A breakdown below BDT 63 with strong volume would invalidate this stance, while a recovery above the BDT 66–67 zone would signal renewed upside momentum. Until either of these scenarios materialize, maintaining a hold position is the most rational and disciplined response to current market conditions.



Momentum indicators further support a neutral stance. The RSI has cooled from overbought levels above 70 to the 40–45 range, indicating that excess bullish momentum has been fully worked off. Importantly, RSI remains above the oversold threshold, suggesting that downside pressure is losing intensity and the stock is stabilizing rather than breaking down. The flattening of the RSI also signals a transition from trending behavior to range-bound price action.

Technically, the current price region around BDT 64–65 represents a confluence zone a former breakout area that now acts as support. The market's ability to hold this level despite multiple tests adds credibility to the consolidation thesis. A sustained move below this zone would weaken the structure, but until that occurs, downside risk appears limited relative to the prior upside achieved.

In investment terms, the chart supports a hold recommendation. The primary uptrend has paused, not reversed, and the stock appears to be digesting earlier gains. For existing investors, exiting at this stage would mean selling into consolidation after momentum has already normalized. For new investors, confirmation of renewed strength such as a volume-backed move above the BDT 67–68 range—would provide a more favorable entry signal.

Overall, BRAC Bank is best characterized as a stock in post-rally stabilization, where patience is warranted. The technical structure favors capital preservation and observation rather than aggressive repositioning, making a hold stance the most prudent course until directional clarity re-emerge

INVESTMENT RATIONALE

BRAC Bank stands out as a high-quality, profitability-led banking franchise in Bangladesh, underpinned by industry-leading ROA (1.46%) and ROE (17.43%), strong revenue and operating profit growth, and a resilient SME- and retail-focused business model. These strengths justify the bank's premium positioning relative to peers and support its long-term earnings compounding

potential. However, valuation discipline tempers near-term upside expectations. At a market price of BDT 63.10, the stock already embeds a significant portion of anticipated forward-earning growth. While forward EPS of BDT 10.67 supports the upper end of our valuation range, trailing P/E and P/B comparisons indicate that the stock is trading above historical and industry-average intrinsic benchmarks, leaving limited margin of safety in the absence of further positive surprises.

From a market behavior perspective, price–volume dynamics and technical indicators suggest post-rally stabilization rather than trend reversal. The stock is holding key support around BDT 63–65, with declining volume on pullbacks and normalized RSI levels, indicating consolidation after a sharp re-rating in 2024–2025. This reinforces the view that the current phase is one of digestion, not deterioration.

FINAL INVESTMENT DECISION: HOLD

Based on the comprehensive fundamental, valuation, price–volume, and technical analyses, we reaffirm a HOLD recommendation on BRAC Bank Limited with a 12-month valuation range of BDT 55–71 per share and a target fair value of BDT 70.7.

BRAC Bank is not a deep-value option, but a quality-at-a-fair-price proposition. Its superior profitability, operational efficiency, and forward growth prospects support long-term ownership, yet its current valuation and technical consolidation argue against aggressive accumulation at present levels.

Accordingly, **HOLD** remains the most balanced and disciplined investment decision at this stage.

STOCK 02: Grameenphone Ltd.

Dhaka Stock Exchange
(DSE: GP)

Sector: Telecommunication
As of 31st Dec 2025

Current Price: BDT 257.90

Recommendation: BUY

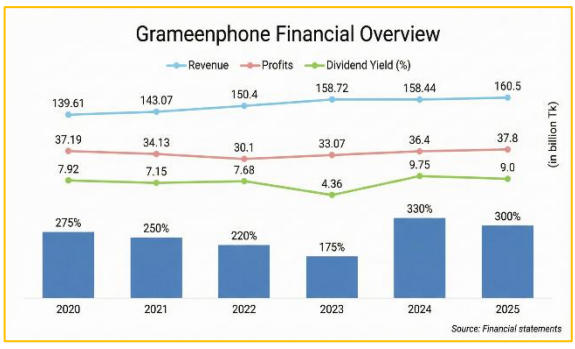
EXECUTIVE SUMMARY: GRAMEENPHONE

VALUATION RATIOS	
P/E Ratio	11.67
EPS	BDT 21.78
Dividend Yield	0.1102
EV/EBITDA	5.46
P/S Ratio	2.19
P/FCF	7.36

Market Capitalization	
Market Cap (in mn)	BDT 343,111
No. of shares outstanding (in mn)	1350.3

PROFITABILITY RATIOS	
ROA	0.1707
ROE	0.5479
Gross Margin	0.7588
Operating Margin	0.3534
Current Ratio	0.19
Debt/Equity	1.54

Grameenphone Ltd. is the largest telecommunications service provider in Bangladesh and a key contributor to the country's digital development. The company commenced its operations on 26 March 1997 and currently serves over 80 million subscribers nationwide. Grameenphone was established through a joint venture between Telenor Group of Norway and Grameen Telecom, an affiliate of the Nobel Prize–winning Grameen Bank, with the objective of making mobile communication accessible to all segments of society, particularly the rural population. As the first GSM-based mobile operator in Bangladesh, Grameenphone introduced modern mobile telecommunication technology to the country and significantly expanded network coverage across both urban and rural areas. Operating under the vision of "Empowering Societies," the company provides a wide range of services, including mobile voice services, mobile data and internet solutions, broadband connectivity, and digital services such as the MyGP app. In addition, Grameenphone offers advanced solutions including IoT and cloud-based services for individual and enterprise customers. Through its strong nationwide network infrastructure, service reliability, and customer-focused approach, Grameenphone plays a vital role in supporting digital inclusion and economic growth in Bangladesh.



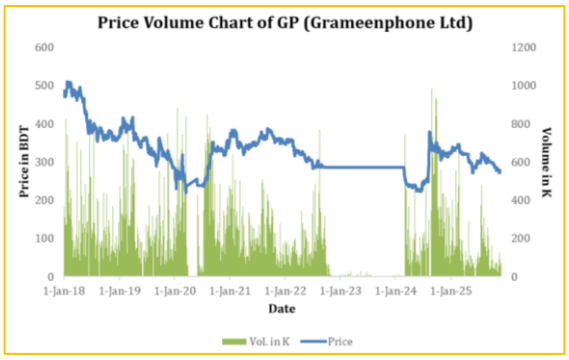
Based on the provided metrics, Grameenphone Ltd. demonstrates robust financial health, supported by a substantial Market Capitalization of BDT 343.11 billion and 1,350.30 million shares outstanding.

Valuation ratios indicate efficient market pricing and strong investor value, featuring a P/E ratio of 11.67x

and an EV/EBITDA of 5.46x. A high EPS of BDT 21.78 combined with an exceptional 11.02% Dividend Yield highlights significant profitability and superior shareholder returns.

Profitability metrics remain elevated, with a Gross Margin of 75.88% and an Operating Margin of 35.34%, reflecting disciplined cost control and operational efficiency. The company's ability to generate returns from its base is evidenced by a ROA of 17.07% and a very high ROE of 54.79%. While the Current Ratio of 0.19 suggests a tight liquidity position, and the Debt/Equity ratio stands at 1.54, the overall data confirms a highly profitable, well-managed entity with strong value characteristics and high investor appeal.

PRICE-VOLUME ANALYSIS



January 2018 – December 2019 In the period from January 2018 to late 2019, GP stock exhibited a distinct bearish trend. The price started at a peak of approximately 500 BDT and experienced a gradual decline, stabilizing around the 300-350 BDT range by the end of 2019. Throughout this phase, trading volume remained high and volatile, indicating active market participation and significant selling pressure as investors adjusted valuations downward from the 2018 highs.

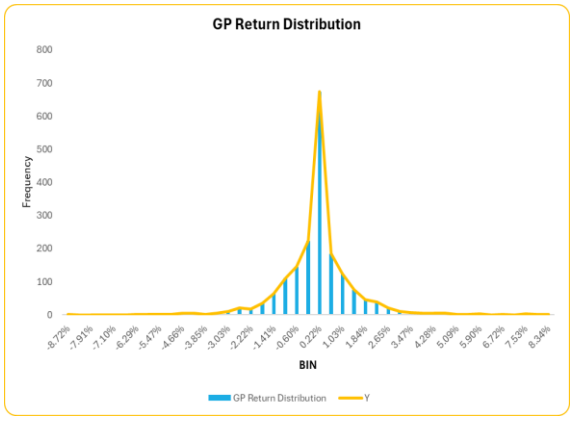
January 2020 – Mid 2021 From early 2020, a steep decline in stock price is visible, bottoming out below 250 BDT. This volatility correlates with the onset of the Covid-19 pandemic, which triggered widespread panic and a sell-off in the capital markets. However, a recovery phase is observable starting mid-2020, where the price rebounded to the 350-400 BDT range. Volume spikes during this recovery suggest renewed investor confidence and opportunistic buying at lower price levels.

Late 2021 – December 2023 Between late 2021 and mid-2022, the stock showed a slow downward trajectory. From September 2022 through practically all of 2023, the chart displays a complete flatline in price at approximately 286 BDT, accompanied by extremely low trading volumes. This phenomenon is directly attributed to the imposition of a "Floor Price" by the Bangladesh Securities and Exchange Commission (BSEC). The regulatory restriction prevented the stock from trading below a set level, effectively freezing liquidity and resulting in minimal trading activity for over a year.

January 2024 – Present (2025) A dramatic shift occurred in early 2024 following the removal of the floor price restriction. The price initially plummeted below 250 BDT as the market corrected itself

to reflect true supply and demand. This period coincides with the highest volume spikes on the chart (reaching nearly 1,000K), indicating massive capitulation and subsequent institutional accumulation. Following this correction, the price staged a sharp rally, briefly reclaiming the 380 BDT level, before entering a new phase of volatility. As of early 2025, the stock shows a stabilizing but slightly bearish trend, trading near the 280-300 BDT range with normalized trading volumes.

RETURN DISTRIBUTION



Grameenphone's return distribution chart highlights the stock's performance characteristics, showing a sharp and tall peak around the 0.22% bin. This high concentration indicates that for most trading days, the stock price moved very little. However, the distribution curve also reveals long tails on both sides, driven by unexpected extreme high and low returns. Specifically, the stock generated a maximum return of 8.75% and a minimum of -8.72%. While the daily movements are typically minimal, the presence of these outliers demonstrates that the stock can generate significant returns or losses in short periods.

VOLATILITY

The return statistics indicate a total range of 17.47%, calculated from the spread between the maximum return of 8.75% and the minimum of -8.72%. This wide dispersion suggests that while stocks are generally stable (as evidenced by the high frequency of low returns), it experiences periods of distinct volatility where price swings are substantial. This 17.47% spread implies that investors faced a non-trivial level of risk, as the stock price fluctuated significantly between its peak and trough performance during the observed period.

SKEWNESS

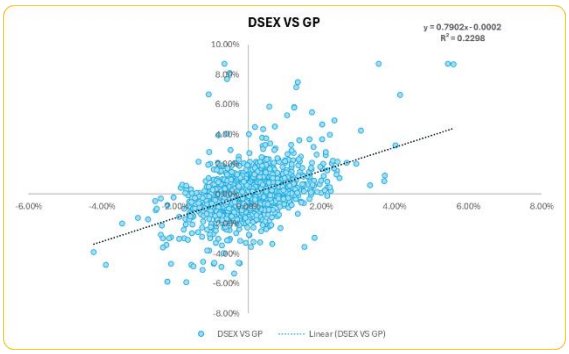
The skewness of GP stock returns is 1.0389, indicating a positive skew. This means the distribution is right-skewed, where the tail on the right side is longer or fatter than the left. While the returns are generally clustered around zero, the frequency or magnitude of large positive returns slightly outweighs that of the extremely negative returns. This asymmetry suggests that, historically, the stock has had a higher propensity for extreme upside outliers compared to downside drops.

KURTOSIS

The kurtosis of the GP return distribution is 7.9914, which indicates a leptokurtic distribution. This value is considerably higher than the normal distribution kurtosis of 3. This high kurtosis signifies

that the stock has "fat tails" and a sharper peak than a normal curve. For investors, this implies that while the stock usually exhibits low volatility, there is a higher probability of rare, extreme events (both positive and negative) occurring than a standard normal model would predict. This characteristic suggests a higher risk profile regarding sudden, large price movements.

INTERPRETATION OF DSEX VS GP



The Beta of GP is derived from the regression equation ($y = 0.7902x - 0.0002$) as 0.7902. This indicates that if the market (DSEX) moves by 1%, GP moves around 0.79% in the same direction. For instance, if the market shows a positive return of 1%, GP is expected to show a positive return of approximately 0.79% and vice versa. Based on this, we can classify GP as a relatively stable stock in terms of market sensitivity (defensive stock), as its movements are less volatile than the market index itself (beta < 1).

The (R^2) value of GP is 0.2298, which demonstrates a weak-to-moderate correlation between the market return and the stock return. From this (R^2) value, it is evident that only around 22.98% of GP's price movements are explained by market performance (systematic risk), while the remaining 77.02% is driven by unsystematic risk. Hence, we can conclude that the systematic risk of GP is low compared to its significantly higher unsystematic risk (company-specific factors such as regulatory changes or operational efficiency).

However, the beta value of 0.7902 derived from this regression analysis may not fully reflect GP's long-term sensitivity. The dataset likely includes volatile periods (such as the 2020 market shocks) which can skew the calculated beta higher than its historical average. For comparison, the beta obtained from Amarstock.com is typically around 0.43, which is notably lower than our calculated figure. This discrepancy suggests that while our specific timeframe captures a higher sensitivity (0.79), longer-term databases may smooth out these anomalies, portraying GP as even less sensitive to market fluctuations than this specific chart suggests.

VALUATION OF GRAMEENPHONE LTD. USING DDM:

Two valuation methods in DDM were used to estimate the intrinsic value of Grameenphone's stock: using the Gordon growth model and using the average of the last 5 years P/E ratio of GP.

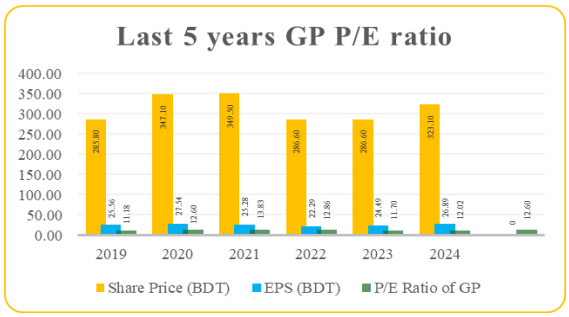
Grameenphone has historically maintained a stable dividend policy, though recent years have shown volatility due to earnings constraints and subsequent recoveries. Following the recovery pattern

observed in 2024, where the company increased payouts to restore investor confidence, the projected cash flows for the upcoming years (Year 1 to Year 5) show a steady upward trend ranging from BDT 30.31 to BDT 34.10. This projection reflects the company's commitment to delivering returns to shareholders and stabilizing its earnings per share (EPS) growth trajectory.

Terminal value using gordon model					**211.72**
Terminal value using last 5 years company PE					**338.24**
Total CF gordon	30.31	31.52	32.79	34.10	238.56
Present value gordon	26.78	24.61	22.62	20.78	128.45
Year	**1**	**2**	**3**	**4**	**5**
Intrinsic value/share	**223.23**	**using gordon**			
Total CF PE	30.31	31.52	32.79	34.10	365.09
Present value PE	26.78	24.61	22.62	20.78	196.57
Intrinsic value/share	**291.35**	**using company PE**			
Price As Of 2 Jan 2026	**BDT 257.90**	**undervalued according to both Gordon and PE**			

To calculate the intrinsic value of GP's stock, we first used the Gordon growth model and found it to be valued at BDT 223.23. Based on this specific calculation, the company appears to be overvalued with the current market price standing at BDT 257.90 as of January 2, 2026. This valuation relies on the terminal value of BDT 211.72 derived from the constant growth assumption.

However, when applying the market-based approach using the last 5 years' company P/E ratio, the intrinsic value is calculated at BDT 291.35. In contrast to the Gordon model, this method suggests the stock is undervalued at the current price level, driven by a higher terminal value estimate of BDT 338.24. This discrepancy suggests that while the strict dividend growth assumptions point to a premium pricing in the market, the company's historical earnings multiples indicate there is still potential upside for investors.



To determine the intrinsic value of Grameenphone's stock using the P/E ratio, we have used the 5-year (2020-2024) average P/E ratio of the company. We believe this is a reliable method for intrinsic value estimation, this helps to capture the company's true, long-term valuation behavior and reduces the risk of mispricing due to temporary or external distortions. The telecom industry operates with vastly different business models, capital structures, and growth rates. Using Grameenphone's own average helps us avoid misevaluation due to irrelevant peer dynamics. Hence the 5-year average P/E ratio of Grameenphone has been calculated to be 12.60 which was used to determine the intrinsic value of its

shares. The intrinsic value was determined to be BDT 291.35, which is higher than the current market trading price of BDT 257.90 (as of January 2, 2026), indicating that the stock is undervalued.

SENSITIVITY ANALYSIS

For the sensitivity analysis of the Gordon Growth Model to estimate the intrinsic value of Grameenphone's stock under varying assumptions of growth rate and required return, we can say that the stock's intrinsic value is highly sensitive to changes in the growth rate assumption. As the growth rate increases from 6.0% to 7.5%, the estimated value decreases significantly (from BDT 478.82 to BDT 385.27 at a 15.5% required return). On the other hand, changes in the required return show a minimal impact on the valuation within this specific range; as the required return increases from 15.5% to 16.8%, the value drops only marginally (e.g., at 6.0% growth, value shifts from 478.819 to 478.810). The sensitivity analysis suggests a potential valuation range between BDT 385.26 and BDT 478.82, which is significantly higher than the base intrinsic value calculation of BDT 223.23.

Gordon Sensitivity Analysis

		Required return					
	223.23	**15.5%**	**15.8%**	**16.0%**	**16.3%**	**16.5%**	**16.8%**
Growth rate	**6.0%**	478.819	478.817	478.815	478.813	478.811	478.810
	6.3%	456.561	456.560	456.558	456.556	456.554	456.552
	6.6%	436.319	436.317	436.316	436.314	436.312	436.310
	6.9%	417.830	417.828	417.826	417.824	417.822	417.821
	7.2%	400.874	400.872	400.870	400.868	400.867	400.865
	7.5%	385.267	385.266	385.264	385.262	385.260	385.259

The sensitivity analysis based on different P/E ratios and required returns show that the intrinsic value of Grameenphone's stock is responsive to changes in both these variables. As the P/E ratio increases from 6.0% to 7.5%, the stock's value decreases—from BDT 134.36 to 129.00 at a 15.5% required return—demonstrating that changes in the earnings multiple assumptions have a notable impact on valuation. Conversely, as the required return increases from 15.5% to 16.8%, the value consistently sees a slight increase across all P/E levels (e.g., at a P/E of 6.6%, the value rises from 132.17 to 132.41). The calculated outcomes vary from BDT 129.00 to BDT 134.61, indicating that the valuation is sensitive to investor sentiment and market return expectations.

Sensitivity PE

		Required return					
	291.35	**15.5%**	**15.8%**	**16.0%**	**16.3%**	**16.5%**	**16.8%**
P/E Ratio	**6.0%**	134.356	134.406	134.456	134.506	134.557	134.607
	6.3%	133.256	133.306	133.355	133.404	133.454	133.503
	6.6%	132.170	132.219	132.268	132.316	132.365	132.414
	6.9%	131.098	131.146	131.194	131.242	131.291	131.339
	7.2%	130.040	130.087	130.135	130.182	130.230	130.277
	7.5%	128.995	129.042	129.089	129.135	129.182	129.229

TECHNICAL ANALYSIS



The 1-year candlestick chart of GP from January 2025 to January 2026 obtained from investing.com shows the fluctuating nature of its stock prices, with the price reaching its lowest at BDT 254.10 in January of this year. The stock price rose to its highest at approximately BDT 335 in the ending period of February 2025. While the stock showed resilience with a recovery rally in July, 2025 where prices briefly touched BDT 320—the decline in prices from August 2025 onwards has been persistent. This downward trend in the latter half of the year suggests a shift in market sentiment, as the stock failed to sustain its mid-year momentum and faced continued selling pressure. Consequently, the final quarter of 2025 and the start of 2026 have been characterized by a bearish trajectory, driving the price down to its current level.

INVESTMENT DECISION

Grameenphone Ltd. remains a dominant market leader with strong fundamentals and stable cash flows. Despite liquidity pressure during the 2022–2023 floor price period, the company rebounded in 2024 with solid profitability, reflected in a high ROE of 54.79%, strong gross margin of 75.88%, and an attractive dividend yield of 11.02%. Its low beta of 0.79 positions the stock as a defensive investment, offering stability during market downturns, though with limited upside in strong bull markets.

Valuation results are mixed: the Gordon Growth Model suggests slight overvaluation, while the 5-year average P/E indicates undervaluation. Overall, the stock appears fairly valued to mildly undervalued, supported by a strong dividend cushion.

Recommendation: Buy. Grameenphone is well-suited for income-focused investors seeking steady returns, low volatility, and reliable dividend income rather than aggressive capital appreciation.



STOCK 03: Square Pharmaceuticals Ltd.

Dhaka Stock Exchange
(DSE: SQPH)

Sector: Bank
As of 2nd Jan 2026

Current Price: BDT 198.30

Recommendation: Buy

EXECUTIVE SUMMARY: SQUARE PHARMACEUTICALS

Market Data	
Market Capitalization (BDT)	178.09 billion
Book Value Per Share)BDT)	170
EPS (BDT)	27.04
Audited P/E	8.93x
Free Float	0.6482
Dividend Yield	5-6%

Established in 1958 and listed on the Dhaka Stock Exchange since 1991, Square Pharmaceuticals Ltd. is the flagship enterprise of the Square Group and the undisputed leader of Bangladesh's pharmaceutical industry. As a premier blue-chip company with a market capitalization often exceeding BDT 180 billion, it specializes in the manufacturing and global distribution of generic medicines across diverse therapeutic segments, including antibiotics, cardiovascular, and herbal products. The company's dominant market position is underpinned by its WHO-GMP compliant facilities, an extensive distribution network, and a reputation for consistent financial performance, making it a cornerstone defensive stock for investors and a growing international exporter to over 40 countries.

FUNDAMENTAL ANALYSIS

Square Pharmaceuticals Ltd. is a leading pharmaceutical company operating in Bangladesh. The company is currently in a mature stage of its business life cycle and is not aggressively pursuing large-scale expansion. As a result, Square Pharmaceuticals retains a reasonable portion of its net profit for operations while distributing a significant share to its shareholders in the form of cash dividends. This dividend-paying tendency reflects the company's stable earnings base and strong cash flow position. The company's average dividend payout ratio, as shown in the DDM sheet, indicates consistent dividend distribution, which supports the use of dividend-based valuation models. Given Square Pharmaceuticals' reliable dividend history, it is expected that the company will continue providing regular cash dividends in the future. Therefore, the Dividend Discount Model (DDM) was used to determine the intrinsic value of the company's stock.

In order to estimate the intrinsic value, we calculated the required rate of return (2.44%) and the net profit growth rate, which was considered as the dividend growth rate under the assumption that dividends grow in line with net profit. According to the DDM sheet, Square Pharmaceuticals exhibits positive and stable net profit growth while maintaining a sustainable dividend payout ratio. Since the dividend growth rate is lower than the required rate of return (G < R), the Gordon Growth Model

was considered appropriate for valuation. Using the Gordon method, the intrinsic value of Square Pharmaceuticals' stock was calculated to be BDT 198.45.

To support the valuation, a relative valuation approach using the industry P/E multiple was also applied. Based on the industry average P/E ratio, the intrinsic value of Square Pharmaceuticals' stock was estimated at BDT 145.30. The current market price of the stock is BDT 198.30, which is lower than both the intrinsic value obtained from the Gordon Growth Model and the industry P/E valuation. This indicates that Square Pharmaceuticals' stock is overvalued in the market.

Terminal value using gordon model					**278.09**
Terminal value using industry PE					**178.30**
Total CF gordon	12.24	13.40	14.68	16.07	295.68
Present value gordon	10.79	10.42	10.06	9.71	157.47
year	**1**	**2**	**3**	**4**	**5**
Intrinsic value/share	**198.45**	**using gordon**			
Total CF PE	12.24	13.40	14.68	16.07	195.90
Present value PE	10.79	10.42	10.06	9.71	104.33
Intrinsic value/share	**145.30**	**using industry PE**			
Current price 31 Jan 2026	**198.30**	**Overvalued**			

Sensitivity Gordon

		Required return					
	198.45	15%	16%	17%	18%	19%	20%
Growth rate	6.0%	154.81	138.91	125.91	115.10	105.96	98.13
	6.5%	161.07	143.74	129.72	118.15	108.44	100.18
	7.0%	168.11	149.10	133.90	121.48	111.13	102.39
	7.5%	176.10	155.10	138.53	125.13	114.06	104.77
	8.0%	185.22	161.84	143.67	129.14	117.25	107.35
	8.5%	195.75	169.49	149.41	133.57	120.75	110.16
	9.0%	208.03	178.23	155.88	138.49	124.59	113.22

Sensitivity PE

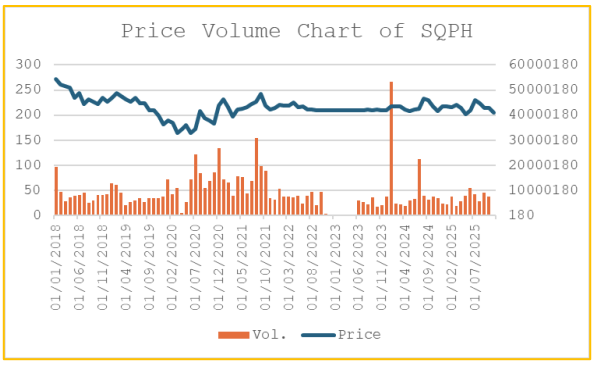
		Required return					
	145.30	15%	16%	17%	18%	19%	20%
Growth rate	20%	50.12	48.84	47.62	46.45	45.32	44.24
	25%	50.55	49.26	48.03	46.84	45.69	44.59
	30%	50.99	49.68	48.43	47.22	46.06	44.94
	35%	51.43	50.10	48.83	47.60	46.43	45.30
	40%	51.87	50.52	49.23	47.99	46.80	45.65
	45%	52.30	50.94	49.63	48.37	47.17	46.01

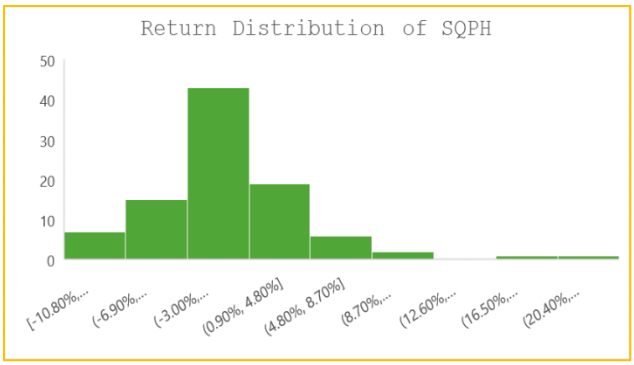
TECHNICAL ANALYSIS



The technical chart of Square Pharmaceuticals Ltd. from Investing.com supports the hypothesis that the stock is undervalued and suitable for buying. The stock experienced a prolonged downtrend from August to early December but later stabilized around the **BDT 200–205** range, which has acted as a strong support level. The current market price of **BDT 204** lies close to this support, indicating limited downside risk. Recent price movements show signs of reversal, with higher lows and improving price action, accompanied by increased trading volume, suggesting weakening selling pressure and growing investor interest. This consolidation, followed by a slight upward movement, signals accumulation, which aligns with the fundamental analysis indicating undervaluation.

The price-volume chart reveals a pattern of steady price consolidation followed by high-volume breakouts, indicating strong institutional accumulation and reinforcing the stock's position as a resilient market leader.

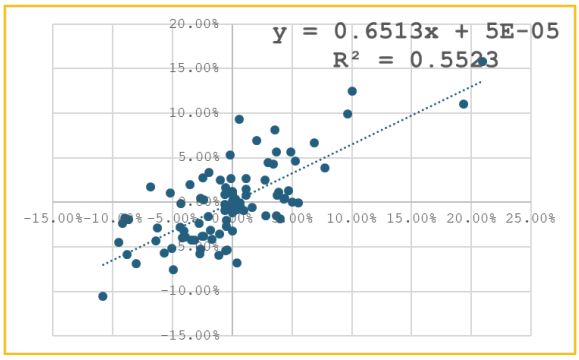




Negative Skewness: This means the "tail" on the left side is longer. In investment terms, it suggests that while most returns are positive, there is a risk of infrequent but large losses.

High Kurtosis: This shows the graph is "peaked" in the middle with thick tails. It tells investors that the stock is more likely to experience extreme price jumps (either up or down) than a typical stable asset would.

BETA ANALYSIS



The estimated beta of Square Pharmaceuticals Ltd. is 0.6513, which is less than 1, indicating that the stock is less volatile than the overall market (DSEX). This means that for a 1% change in the market, Square Pharma's return changes by approximately 0.65%, reflecting lower market sensitivity. Such a beta is appropriate for the defensive pharmaceutical sector, where demand remains stable across economic cycles. As a blue-chip company with strong fundamentals, stable earnings, and consistent dividends, Square Pharmaceuticals' lower risk profile is justified. Additionally, the R^2 value of 0.5523 shows a moderate to strong relationship with market returns, making the beta statistically reliable.

FINAL VERDICT

The stock's current market price as of 1st Jan 198 is higher than its intrinsic value using valuation approaches based on the Industry P/E valuation i.e. BDT 145.30

This indicates that the stock is overvalued in the market. The company's stable dividend payout ratio, consistent net profit growth, and stable required rate of return (13.43%) reflect strong fundamentals and marginal risks. Additionally, the ability of Square Pharmaceuticals to sustain dividend payments supports long-term investor confidence.

Therefore, considering both absolute and relative valuation outcomes, along with the company's strong financial position, Square Pharmaceuticals Ltd. presents an attractive investment opportunity, and investors are recommended to **buy** the stock at the current market price



EXECUTIVE SUMMARY: APEX TANNERY LTD.

Market Data	
Market Capitalization (BDT)	870,000,000
Shareholder's Equity (BDT)	835,900,000
Book Value Per Share (BDT)	30.19
Forward P/E	n/a
EPS (BDT)	(14.63)
Audited P/E	3.90
Free Float	72.13%
Last Year's Revenue Growth	-23.70%
Operating Profit Growth	-17.34%

Apex Tannery Limited is a leading leather manufacturing company based in Bangladesh. It is part of the Apex Group, one of the country's most prominent footwear and leather exporters. The company specializes in producing high-quality finished leather for footwear, leather goods, and accessories. Apex Tannery operates with modern machinery and follows international quality standards. It places strong emphasis on environmental compliance and sustainable production practices. Its products are exported to various international markets, supporting Bangladesh's leather industry.

FUNDAMENTAL ANALYSIS

Apex Tannery LTD is a tannery company in Bangladesh the company has been giving dividend for a long time. The percentages of cash dividends are also quite small 10%, 5% and 5% in 2022, 2023 and 2024 respectively. However, we assume that the company will have ability to give a bigger percentage of cash dividends in near future and that is why we are determining the company's stock by Free Cash Flow to Equity (FCFE) Model.

To do so, we calculated both the Growth Rate of FCFE (9.62%) and Net Profit Growth Rate (20.57%). Assuming FCF of the company will grow at the rate of net profit growth over the years, we forecast FCF for the next five years and then use a multiplier model for terminal value.

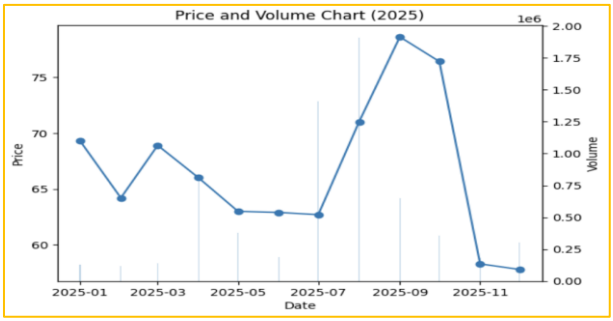
Forecast	FCFE	PV
2025	535,862,387.24	488,854,176.81
2026	541,572,319.73	450,721,806.58
2027	547,343,094.94	415,563,897.30
2028	553,175,361.20	383,148,430.41
2029	559,069,773.72	353,261,485.60
Terminal Value calculated using last 5 years PE		
2022	48913014503	30,906,847,375
Total		32,998,397,172
divide by no of shares		15,240,000.00
Intrinsic value		2,165.25
3rd January Price 2026		59.5

As we can see, the intrinsic value is 2165 taka and the market price is 59.50 taka, this means that the investors who already have the stock should hold onto it and investors can buy the stock as the stock price is expected to increase.

TECHNICAL ANALYSIS

The RSI (Relative Strength Index) for apex tannery is 46.4 according to trading view Bangladesh (n.a).





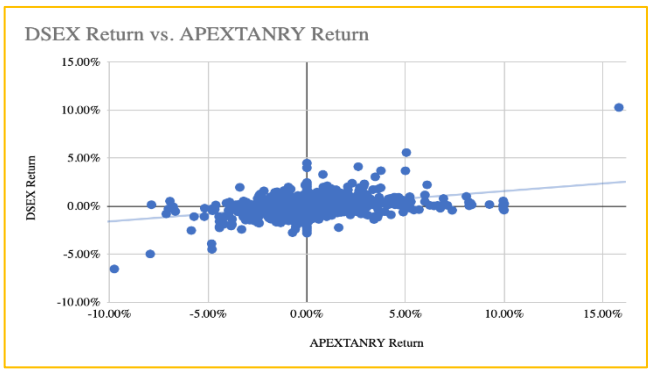
The price volume chart at the beginning of October shows that the price is increasing and it has the potential to increase even further, which justifies our intrinsic value and our calculation through the FCFE model.

KURTOSIS

For apex tannery a kurtosis of 8.89 demonstrates that the distribution of values is highly leptokurtic, i.e., the distribution is sharp-peaked and has very heavy tails. This implies that extreme values or outliers are very frequent in comparison to normal distribution. This indicates an increase in tail risk, and rare but large movements are more frequent.

SKEWNESS

The skewness of 1.04 for APEX TANNERY in a skewed distribution implies the right tail to be longer than the left tail, that is, the shape is positively (right) skewed. This implies that there is a greater frequency of higher extreme values than low extremes. The bulk of the observations is located on the lower part of the distribution and there are large upward values occasionally.



For Apex Tannery the beta is 0.87 which means it is less volatile than the market. The beta of 0.87 shows that Apex Tannery (APEXTANRY) is theoretically less volatile as compared to the entire Bangladesh stock market (DSE). It implies that the stock market is 13 percent less probable to fluctuations in the market; in case the market index increases or decreases by 10 percent, the price of Apex Tannery would generally change by 8.7 percent. This qualifies the stock as a defensive investment, which provides the shareholders with a rougher ride during turbulent times in the market.

PART 02: PORTFOLIO ANALYSIS

PERFORMANCE OF STOCK

	APEX Tannery Ltd	BRAC Bank	GrameenPhone	Square Pharma
Daily Variance	0.02%	0.01%	0.01%	0.03%
Daily Stdev	1.32%	1.11%	0.77%	1.86%
No of trading days	250	250	250	250
Annualized variance	**4.38%**	**3.06%**	**1.47%**	**8.68%**
Annualized Stdev	**20.92%**	**17.49%**	**12.11%**	**29.46%**
Daily Average Return	**0.03%**	**0.22%**	**0.11%**	**0.31%**
Annualized return	**7.65%**	**53.93%**	**28.30%**	**77.03%**

Covariance

	APEX Tannery L	BRAC Bank	GrameenPhone	Square Pharma
APEX Tannery Ltd	0.000175	0.000024	-0.000007	-0.000002
BRAC Bank	0.000024	0.000126	0.000036	0.000027
GrameenPhone	-0.000007	0.000036	0.000060	0.000020
Square Pharma	-0.000002	0.000027	0.000020	0.000360

Risk Profile (Volatility Analysis)

- Grameenphone emerges as the least volatile stock in the group. With the lowest daily and annualized standard deviation, it demonstrates highly stable price behavior. This indicates strong defensive characteristics, making it suitable for conservative and income-oriented investors who prioritize capital preservation and lower uncertainty.
- BRAC Bank exhibits moderate volatility, lower than most equities in emerging markets. Its risk level reflects relatively stable price movements, likely supported by consistent banking operations and earnings visibility. This places BRAC Bank in the category of balanced risk investments.
- APEX Tannery Ltd shows moderately high volatility, indicating greater sensitivity to market movements and sector-specific factors. While not excessively risky, its price fluctuations are meaningfully higher than those of BRAC Bank and Grameenphone, suggesting increased uncertainty.
- Square Pharmaceuticals has the highest volatility among the four stocks. Its elevated daily and annualized standard deviation signals significant price swings, implying higher market expectations, strong growth dynamics, and sensitivity to investor sentiment. This level of volatility increases both the potential and downside risk.

Volatility ranking (Lowest to Highest):

1. Grameenphone
2. BRAC Bank
3. APEX Tannery Ltd
4. Square Pharmaceuticals

Return Performance

- Square Pharmaceuticals delivers the strongest return performance, with the highest daily average return translating into the highest annualized return. This indicates strong price appreciation momentum and investor confidence, justifying its higher volatility. The stock is clearly positioned as a growth-oriented investment.
- BRAC Bank demonstrates impressive return generation, with a high annualized return relative to its moderate risk level. This suggests effective risk compensation and reflects strong fundamentals, growth in earnings, and market re-rating potential.
- Grameenphone produces moderate but stable returns. While its annualized return is lower than BRAC Bank and Square Pharmaceuticals, it remains attractive when viewed in conjunction with its very low volatility. The stock emphasizes consistency rather than aggressive growth.
- APEX Tannery Ltd has the lowest return profile among the four stocks. Although returns are positive, they are comparatively modest, indicating weaker price momentum and a less favorable return outlook relative to the level of risk assumed.

Risk–Return Trade-Off

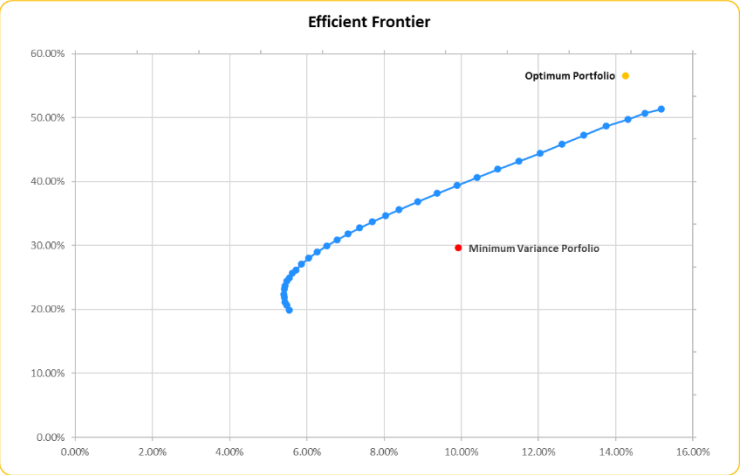
A comparison of returns relative to volatility highlights investment efficiency:

- BRAC Bank offers the most balanced risk–return profile, delivering strong returns without excessive volatility. This makes it attractive for investors seeking growth with controlled risk.
- Grameenphone provides low risk with stable returns, positioning it as a defensive stock suitable for conservative portfolios and dividend-focused strategies.
- Square Pharmaceuticals represents a high-risk, high-return profile, suitable for aggressive investors with higher risk tolerance and a longer investment horizon.
- APEX Tannery Ltd displays a less efficient risk–return combination, where volatility is relatively high compared to the returns generated.

Overall Assessment

The stock performance analysis highlights clear differentiation in risk and returns characteristics across the selected equities. BRAC Bank and Square Pharmaceuticals stand out in terms of return generation, while Grameenphone excels in stability. APEX Tannery Ltd, although positive, underperforms relative to peers on a risk-adjusted basis. Together, these stocks offer diversification opportunities across defensive, balanced, and growth-oriented investment strategies.

EFFICIENT FRONTIER ANALYSIS



The efficient frontier is constructed using 34 different portfolio weight combinations, each representing a unique risk–return outcome derived from the same asset universe. The resulting curve shows a well-defined, convex frontier, indicating effective diversification benefits across portfolios.

The Minimum Variance Portfolio delivers the lowest achievable risk, while portfolios along the upward-sloping segment dominate inferior combinations by offering higher expected returns for incremental increases in volatility. The Optimum Portfolio, located toward the upper end of the frontier, represents the best risk-adjusted outcome, where return gains are most efficiently compensated for risk.

In terms of weights distribution, lower-risk portfolios are driven by more balanced and defensive weight allocations, limiting exposure to high-volatile assets. As expected, return targets increase, portfolio weights progressively tilt toward higher-return (and higher-risk) assets, leading to greater concentration and higher overall volatility. This transition highlights the trade-off between diversification and returns maximization.

Overall, the frontier confirms that portfolio performance improves materially through optimized weighting, rather than equal or ad-hoc allocations, allowing investors to select portfolios aligned with their risk tolerance.

VARYING WEIGHT ALLOCATED PORTFOLIO

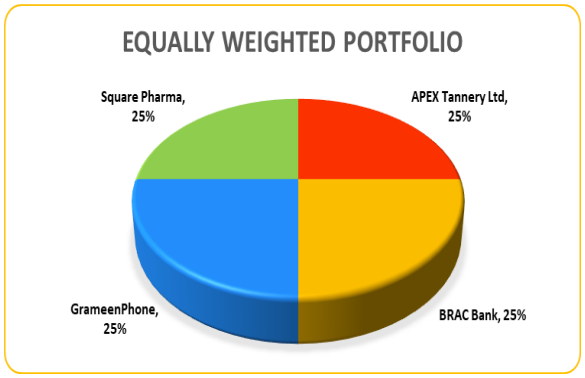
Portfolio 1	APEX Tannery Ltd	BRAC Bank	GrameenPhone	Square Pharma
Equally Weighted	25%	25%	25%	25%
Portfolio return	42.72%			
Portfolio Risk	11.98%			
Sharpe ratio	2.93			

Portfolio 2	APEX Tannery Ltd	BRAC Bank	GrameenPhone	Square Pharma
Random no	0.7461	0.4853	0.7544	0.3092
Random weight	33%	21%	33%	13%
Portfolio return	34.35%			
Portfolio Risk	10.86%			
Sharpe ratio	2.46			

Portfolio 3	APEX Tannery Ltd	BRAC Bank	GrameenPhone	Square Pharma
Minimum Variance	24%	8%	61%	7%
Portfolio return	29.65%			
Portfolio Risk	9.92%			
Sharpe ratio	2.22			

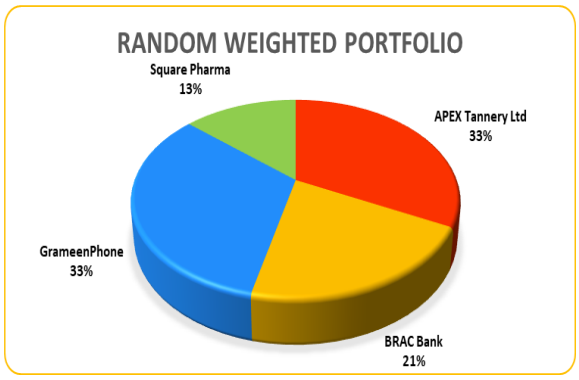
Portfolio 4	APEX Tannery Ltd	BRAC Bank	GrameenPhone	Square Pharma
Optimum Portfolio	0%	51%	21%	28%
Portfolio return	56.59%			
Portfolio Risk	14.25%			
Sharpe ratio	3.44			

Portfolio 1: Equally Weighted Portfolio



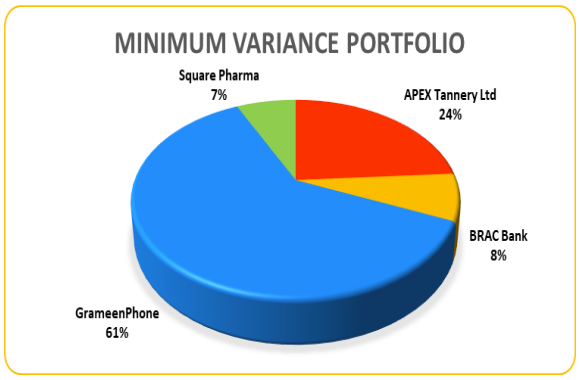
The equally weighted portfolio assigns 25% to each stock, providing maximum diversification neutrality. It delivers a strong return of 42.72% with moderate risk (11.98%), resulting in a Sharpe ratio of 2.93. This indicates efficient diversification, though the portfolio does not fully exploit differences in individual asset risk–return characteristics.

Portfolio 2: Randomly Weighted Portfolio



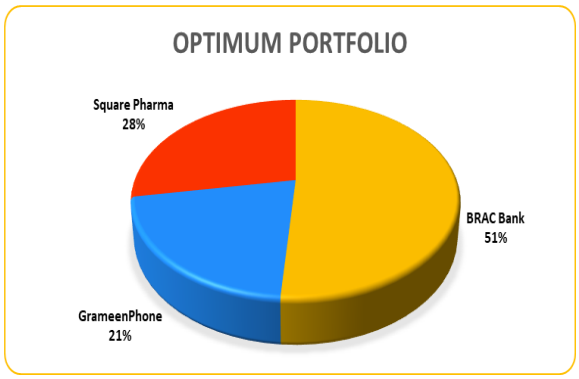
Using uneven, randomly generated weights, this portfolio shows a lower return of 34.35% with slightly reduced risk (10.86%), producing a Sharpe ratio of 2.46. The decline in risk-adjusted performance highlights that non-optimized weight dispersion can dilute efficiency, even when risk is marginally reduced.

Portfolio 3: Minimum Variance Portfolio



The minimum variance portfolio is heavily skewed toward Grameenphone (61%), with limited exposure to higher-volatility stocks. This structure achieves the lowest risk (9.92%) but also the lowest return (29.65%), resulting in a Sharpe ratio of 2.22. It is best suited for risk-averse investors, prioritizing stability overgrowth.

Portfolio 4: Optimum (Maximum Sharpe) Portfolio



The optimum portfolio reallocates weights toward BRAC Bank (51%) and Square Pharma (28%), while eliminating exposure to APEX Tannery. This concentrated allocation delivers the highest return (56.59%) with higher risk (14.25%) but crucially achieves the highest Sharpe ratio (3.44). This confirms it as the most risk-efficient portfolio among the four.

Portfolio Risk–Return Analysis

Across the four portfolios, a clear risk–return trade-off emerges as asset weights vary.

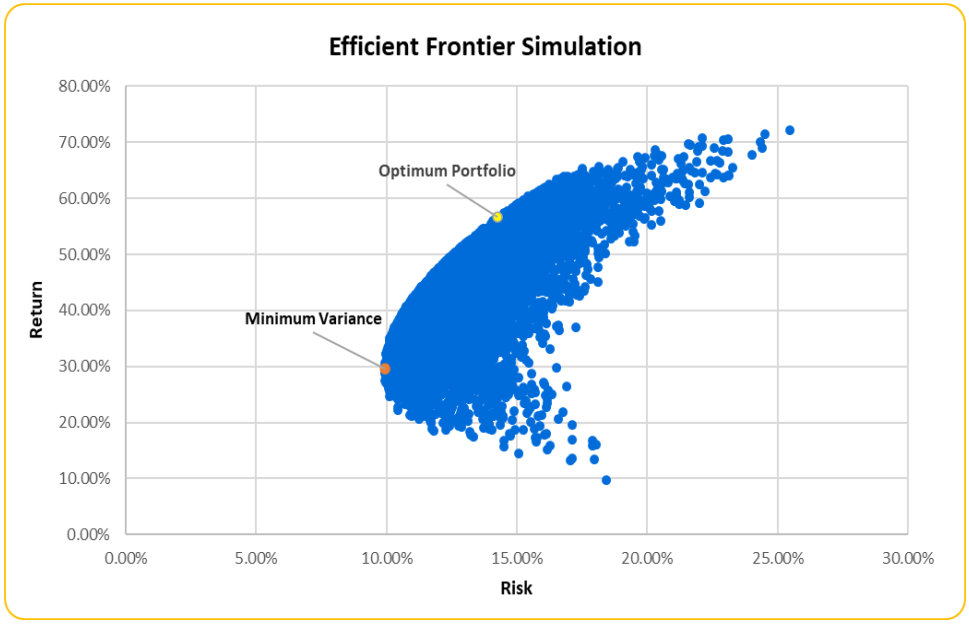
- The Minimum Variance Portfolio exhibits the lowest risk (9.92%) but also the lowest return (29.65%), reflecting its defensive, stability-focused allocation.
- The Randomly Weighted Portfolio slightly reduces risk compared to the equal-weighted case, but this comes at the cost of a notably lower return (34.35%), indicating suboptimal risk–return efficiency.

- The Equally Weighted Portfolio strikes a balanced middle ground, delivering a strong 42.72% return at moderate risk (11.98%), supported by broad diversification.
- The Optimum Portfolio carries the highest risk (14.25%), but this is compensated for by a substantially higher return (56.59%), making it the most attractive on a risk-adjusted basis.

Overall, as portfolios move from defensive to optimized allocations, risk increases, but returns rise more than proportionally, confirming that efficient weighting improves the risk–return profile rather than diversification alone.

EFFICIENT FRONTIER SIMULATION

The efficient frontier simulation illustrates the full opportunity set of portfolios generated through many randomized weight combinations across the selected assets. Each point represents a feasible portfolio, mapping its expected return against corresponding risk, and collectively forming a dense, curved cloud that converges into a clear efficient boundary.



At the left-most edge of the distribution lies the minimum variance portfolio, where risk is minimized at roughly the 10% level and expected returns cluster around the high-20% to low-30% range. This region reflects portfolios dominated by defensive allocations, where diversification benefits are maximized but upside potential remains limited. Portfolios below this point are inefficient, as they deliver lower returns without reducing risk further. As the simulation progresses upward and to the right, the portfolio set transitions into the efficient frontier, where each incremental increase in risk is rewarded with a proportionate rise in expected return. The curvature of the frontier highlights diminishing marginal returns at higher risk levels, signaling that beyond a certain point, additional volatility delivers smaller gains in performance.

The optimum portfolio, highlighted near the upper-middle section of the frontier, represents the most efficient risk–return trade-off. Positioned at a moderate risk level with returns exceeding 55%, this portfolio achieves maximum risk-adjusted performance, balancing growth-oriented exposure without excessive volatility. It is within this region that rational investors would concentrate their choices, depending on individual risk tolerance.

Overall, the simulation confirms that portfolio outcomes vary widely under random weighting, but only a small subset of combinations lie on the efficient frontier. The analysis underscores the importance of optimization, showing that superior portfolio performance is not accidental but the result of disciplined weight selection that exploits diversification while preserving return efficiency.

PORTFOLIO SELECTION

Based on the combined risk–return analysis, Sharpe ratios, efficient frontier results, and simulation outcomes, the Optimum Portfolio (Maximum Sharpe Ratio Portfolio) clearly emerges as the best portfolio selection.

This portfolio delivers the highest risk-adjusted performance, with a return of 56.59%, volatility of 14.25%, and a Sharpe ratio of 3.44, outperforming all other portfolio configurations. Its allocation—concentrated primarily in BRAC Bank (51%) and Square Pharmaceuticals (28%), with supportive exposure to Grameenphone (21%) and no allocation to APEX Tannery Ltd—efficiently balances growth potential and diversification.

The optimum portfolio lies on the upper segment of the efficient frontier, where returns are maximized for each unit of risk taken. While it carries higher absolute risk than the minimum variance and equally weighted portfolios, the incremental risk is more than compensated by disproportionately higher returns, making it the most rational choice for investors with a moderate-to-high risk tolerance and a medium- to long-term investment horizon.

In contrast, the Minimum Variance Portfolio is more suitable only for highly conservative investors focused on capital preservation, while the equally weighted and randomly weighted portfolios remain suboptimal due to lower risk-adjusted efficiency.

CONCLUSION

For investors seeking the most efficient use of capital and superior long-term performance, the Optimum (Maximum Sharpe) Portfolio is the preferred and best portfolio selection.

APPENDIX A: Brac Bank

	From Wall Street Journal				From Audtied Statement			
	P/E	P/B	EV/S	EV/EBIDTA	Revenue Growth	Operating Profit Growth	ROA	ROE
BRAC BANK	7.89	1.22	1.25	5.99	39.99%	68.18%	1.46%	17.43%
Jamuna Bank	6.19	0.71	0.53	2.88	35.63%	29.81%	0.83%	13.38%
Mutual Trust Bank	3.82	0.43	0.94	7.76	28.73%	43.70%	0.77%	12.77%
Prime Bank	3.56	0.74	1.8	7.45	35.10%	51.86%	1.46%	20.56%
City Bank	2.8	0.65	0.87	3.89	42.10%	70.79%	1.70%	26.10%
Trust Bank	5.35	0.61	1.27	7.98	-38.02%	-90.95%	0.75%	15.82%
One Bank	7.08	0.30	1.52	32.46	42.13%	117.19%	0.39%	5.54%
Industry	6.628	0.678	1.460	7.30	38.99%	58.63%	1.54%	15.99%

BRAC BANK is trading at a higher PE ratio than industry average but also has a much higher ROA and ROE than industry average

Return Statistics	
Max	29.62%
Min	-21.38%
Mean (Average Daily Return)	0.03%
Median	0.00%

Risk Metrics	
Variance	0.05%
Standard Deviation	2.17%

Distribution Characteristics	
Skewness	1.62
Kurtosis	33.56

Trading Days	
Total Trading Days	1385.00
Annual Trading Days	198.00

Annualized Statistics	
Annualized Average Daily Retu	5.48%
Annualized Variance	9.77%
Annualized Standard Deviatior	31.26%

P/E Based Intrinsic Value

Industry P/E	6.63	
BRAC BANK EPS	6.86	From Audited Statement
Intrinsic value	45.46	
Price, 31st Dec 2025	63.1	From DSE

BRAC BANK is trading at a 42.83% premiumto fair value, so we should sell/short sell

P/B Based Intrinsic Value

Industry P/B	0.68	
BRAC BANK Book Value Per Share	39.21	From Audited Statement
Intrinsic value	26.56	
Price, 31st Dec 2025	63.1	From DSE

BRAC BANK is trading at a 137.53% premium to fair value, however BRAC BANK's competitive advantages is not fully reflected in book value

Industry Average P/E Ratio	6.63
From Wall Street Journal	
Audited EPS	6.86
TTM EPS	7.89
Forward EPS	10.67

BRAC Bank Stock Price Sensitivity Analysis (P/E Based)

		Industry Average P/E Ratio			
		5.00	6.63	7.00	8.00
Audited EPS	6.86	45.46	45.46	45.46	45.46
TTM EPS	7.89	39.45	52.29	55.23	63.12
Forward EPS	10.67	53.35	70.72	74.69	85.36

Industry Average P/B Ratio	0.68
From Wall Street Journal	
Audited BVPS	39.21
TTM BVPS	51.73

BRAC Bank Stock Price Sensitivity Analysis (P/B Based)

		Industry Average P/B Ratio			
		0.68	1.00	1.50	2.00
Audited BVPS	39.21	26.56	39.21	58.82	78.42
TTM BVPS	51.73	35.05	51.73	77.60	103.46

Regression Statistics	
Multiple R	0.445325057
R Square	0.198314407
Adjusted R Square	0.197706148
Standard Error	0.021988181
Observations	1320

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	0.157632	0.157632	326.0360301	2.6952E-65
Residual	1318	0.637227	0.000483		
Total	1319	0.794859			

	Coefficients	*andard Err*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	0.0005	0.0006	0.7847	0.4328	-0.0007	0.0017	-0.0007	0.0017
X Variable 1	0.9911	0.0549	18.0565	0.0000	0.8834	1.0988	0.8834	1.0988

Regression Equation	
Slope, β	0.99
Intercept, α	0.0005
Regression	*0.0005 + 0.9911x*

APPENDIX B: Grameenphone

Source: Bangladesh bank Annual Report

Whole period	*using mkt cap*	*using index*	*using index without outliers*	*Using index*		*Using mkt cap*	
R_m Market Return	18.43%	5.13%	-1.61%	*last 5 years*	last 10 years	last 5 years	last 10 years
Average	13.92%			3.22%	0.70%	10.62%	9.41%

Calculate Beta for the stock and use CAPM and R_m to calculate required rate of return for each stock

Required Return	Rf+Beta(Rm-Rf)
	13.18%

Rf (Treasury Bill Rate 365 days)	10.71%
Rm(to be used in calculation)	13.92%
Beta of GP	0.77

Terminal value using gordon model					**211.72**
Terminal value using last 5 years company PE					**338.24**
Total CF gordon	30.31	31.52	32.79	34.10	238.56
Present value gordon	26.78	24.61	22.62	20.78	128.45
Year	**1**	**2**	**3**	**4**	**5**
Intrinsic value/share	**223.23**	**using gordon**			
Total CF PE	30.31	31.52	32.79	34.10	365.09
Present value PE	26.78	24.61	22.62	20.78	196.57
Intrinsic value/share	**291.35**	**using company PE**			
Price As Of 2 Jan 2026	**BDT 257.90**	**undervalued according to both Gordon and PE**			

Year	Share Price (BDT)	EPS (BDT)	P/E Ratio of GP
2019	285.80	25.56	**11.18**
2020	347.10	27.54	**12.60**
2021	349.50	25.28	**13.83**
2022	286.60	22.29	**12.86**
2023	286.60	24.49	**11.70**
2024	323.10	26.89	**12.02**
Last 5 years GP P/E ratio			**12.60**

Required return	**13.18%**
Beta	**0.77**
Risk free rate	**10.71%**
Market return	**13.92%**
Long term growth rate	**6.5%**
Last 5 years company PE	**12.60**

Peer companies	PE
Robi Axiata	21.19
Bangladesh Submarine Cable	11.55
Adn Tele	22.67
Agni Systems	25.56
Industry PE	**20.24**

APPENDIX C: Square Pharmaceuticals Ltd.

Source: Bangladesh bank Annual Report

Whole period	using mkt cap	*using index*	*using index without outliers*	*Using index*		*Using mkt cap*	
R_m Market Return	18.43%	5.13%	-1.61%	*last 5 years*	last 10 years	last 5 years	last 10 years
Average	13.92%			3.22%	0.70%	10.62%	9.41%

Calculate Beta for the stock and use CAPM and R_m to calculate required rate of return for each stock

Required	Rf+Beta(Rm-Rf)	Rf (Treasury Bill Rate 365 days)	10.71%
	13.43%	Rm(to be used in calculation)	13.92%
		Beta of SQPH	0.8479

Terminal value using gordon model					278.09
Terminal value using industry PE					178.30
Total CF gordon	12.24	13.40	14.68	16.07	295.68
Present value gordon	10.79	10.42	10.06	9.71	157.47
year	1	2	3	4	5
Intrinsic value/share	198.45	using gordon			
Total CF PE	12.24	13.40	14.68	16.07	195.90
Present value PE	10.79	10.42	10.06	9.71	104.33
Intrinsic value/share	145.30	using industry PE			
Current price 31 Jan 2026	198.30	Overvalued			

Required return	13.43%
Beta	0.8479
Market return	13.92%
Risk free rate	10.87%
Long term growth rate	6.50%

Year	PE	Peer companies	PE
2021	17.99	Beximco	13.07
2022	20.51	Renata	19.36
2023	21.41	Beacon	4.1
2024	23.61	Orion	4
2025	27.04		
Average SQUARE PE	22.112	Industry PE	10.1325

APPENDIX D: Apex Tannery Ltd.

Year	Cash from operating	Capex	Net borrowing	FCFE	Growth	
FY 2019	(131,886,136)	(67,002,678)	328,544,385	263,660,927		
FY 2020	(50,937,337)	(3,571,254)	264,460,248	217,094,165	-17.66%	82%
FY 2021	108,711,290	(3,143,200)	58,994,949	170,849,439	-21.30%	79%
FY 2022	100,864,334	(4,565,930)	82,018,369	187,448,633	-13.66%	86%
FY2023	81,983,566	(2,032,612)	57,613,968	141,630,146	-24.44%	76%
FY2024	28,483,861	(154,975,300)	346,753,495	530,212,656	274.36%	374%
						9.62%

Ooperating income	FCFE	Growth	
125,629,365	521,176,428		
90,083,142	358,114,644	-31.29%	69%
148,158,842	210,296,991	-41.28%	59%
150,981,583	237,565,882	12.97%	113%
(6,702,152)	52,944,428	-77.71%	22%
47,812,949	549,541,744	937.96%	1038%
			1.07%

Forecast	FCFE	PV
2025	535,862,387	488,854,177
2026	541,572,320	450,721,807
2027	547,343,095	415,563,897
2028	553,175,361	383,148,430
2029	559,069,774	353,261,486
Terminal Value calculated using last 5 years PE		
2022	**48,913,014,502.59**	30,906,847,375.26
Total		**32,998,397,171.96**
divide by no of shares		15,240,000
Intrinsic value		**2165.25**
3rd January Price 2026		**59.50**

Year	PE
2017	5153.00%
2018	5818.00%
2019	9956.00%
2020	-8965.00%
2021	31783.00%
Average	8749.00%

APPENDIX E: Portfolio Analysis

	APEX Tannery Ltd	BRAC Bank	GrameenPhone	Square Pharma
Daily Variance	0.02%	0.01%	0.01%	0.03%
Daily Stdev	1.32%	1.11%	0.77%	1.86%
No of trading days	250	250	250	250
Annualized variance	4.38%	3.06%	1.47%	8.68%
Annualized Stdev	20.92%	17.49%	12.11%	29.46%
Daily Average Return	0.03%	0.22%	0.11%	0.31%
Annualized return	7.65%	53.93%	28.30%	77.03%

Covariance

	APEX Tannery L	BRAC Bank	GrameenPhone	Square Pharma
APEX Tannery Ltd	0.000175	0.000024	-0.000007	-0.000002
BRAC Bank	0.000024	0.000126	0.000036	0.000027
GrameenPhone	-0.000007	0.000036	0.000060	0.000020
Square Pharma	-0.000002	0.000027	0.000020	0.000360

	APEX Tannery Ltd Weight	BRAC Bank Weight	GrameenPhone Weight	Square Pharma Weight		E(R)	Risk
Combination 1	30%	30%	40%	0%	100%	29.80%	0.69%
Combination 2	29%	30%	39%	2%	100%	30.98%	0.69%
Combination 3	29%	29%	38%	4%	100%	31.70%	0.68%
Combination 4	28%	29%	37%	6%	100%	32.88%	0.68%
Combination 5	28%	28%	36%	8%	100%	33.59%	0.68%
Combination 6	27%	28%	35%	10%	100%	34.78%	0.68%
Combination 7	27%	27%	34%	12%	100%	35.49%	0.68%
Combination 8	26%	27%	33%	14%	100%	36.67%	0.69%
Combination 9	26%	26%	32%	16%	100%	37.39%	0.69%
Combination 10	25%	26%	31%	18%	100%	38.57%	0.70%
Combination 11	25%	25%	30%	20%	100%	39.29%	0.71%
Combination 12	24%	24%	29%	23%	100%	40.70%	0.73%
Combination 13	23%	23%	28%	26%	100%	42.12%	0.76%
Combination 14	22%	22%	27%	29%	100%	43.53%	0.78%
Combination 15	21%	21%	26%	32%	100%	44.94%	0.81%
Combination 16	20%	20%	25%	35%	100%	46.35%	0.85%
Combination 17	19%	19%	24%	38%	100%	47.76%	0.88%
Combination 18	18%	18%	23%	41%	100%	49.18%	0.92%
Combination 19	17%	17%	22%	44%	100%	50.59%	0.96%
Combination 20	16%	16%	21%	47%	100%	52.00%	1.00%
Combination 21	15%	15%	20%	50%	100%	53.41%	1.05%
Combination 22	14%	14%	18%	54%	100%	55.31%	1.11%
Combination 23	13%	13%	16%	58%	100%	57.21%	1.17%
Combination 24	12%	12%	14%	62%	100%	59.11%	1.24%
Combination 25	11%	11%	12%	66%	100%	61.01%	1.30%
Combination 26	10%	10%	10%	70%	100%	62.91%	1.37%
Combination 27	9%	9%	8%	74%	100%	64.81%	1.44%
Combination 28	8%	8%	6%	78%	100%	66.71%	1.51%
Combination 29	6%	7%	5%	82%	100%	68.81%	1.58%
Combination 30	4%	6%	4%	86%	100%	70.92%	1.65%
Combination 31	2%	5%	3%	90%	100%	73.02%	1.72%
Combination 32	1%	3%	2%	94%	100%	74.67%	1.79%
Combination 33	0%	2%	1%	97%	100%	76.08%	1.84%
Combination 34	0%	0%	0%	100%	100%	77.03%	1.90%

	APEX Tannery Ltd	BRAC Bank	GrameenPhone	Square Pharma
Daily Variance	0.02%	0.01%	0.01%	0.04%
Daily Stdev	1.32%	1.12%	0.77%	1.90%
No of trading days	250	250	250	250
Annualized variance	4.37%	3.16%	1.50%	8.99%
Annualized Stdev	20.90%	17.76%	12.25%	29.99%
Daily Average Return	0.03%	0.22%	0.12%	0.31%
Annualized return	6.80%	55.69%	29.68%	78.70%

Portfolio 1	APEX Tannery Ltd	BRAC Bank	GrameenPhone	Square Pharma
Equally Weighted	25%	25%	25%	25%
Portfolio return	42.72%			
Portfolio Risk	11.98%			
Sharpe ratio	2.93			

Portfolio 2	APEX Tannery Ltd	BRAC Bank	GrameenPhone	Square Pharma
Random no	0.5667	0.8844	0.6545	0.7450
Random weight	20%	31%	23%	26%
Portfolio return	46.01%			
Portfolio Risk	12.34%			
Sharpe ratio	3.11			

Portfolio 3	APEX Tannery Ltd	BRAC Bank	GrameenPhone	Square Pharma
Minimum Variance	24%	8%	61%	7%
Portfolio return	29.65%			
Portfolio Risk	9.92%			
Sharpe ratio	2.22			

Portfolio 4	APEX Tannery Ltd	BRAC Bank	GrameenPhone	Square Pharma
Optimum Portfolio	0%	51%	21%	28%
Portfolio return	56.59%			
Portfolio Risk	14.25%			
Sharpe ratio	3.44			